# Kernix 0.9.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

**25** workloads

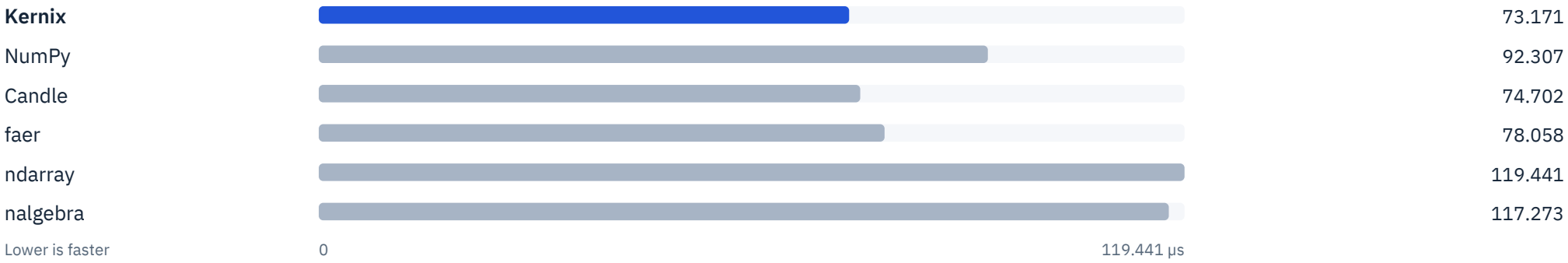
**100** observations

Source precision retained

## Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product




## How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

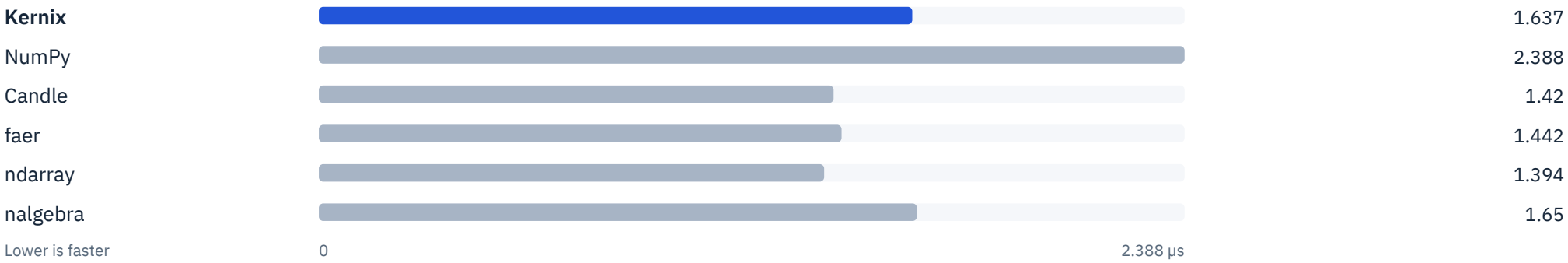
# Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

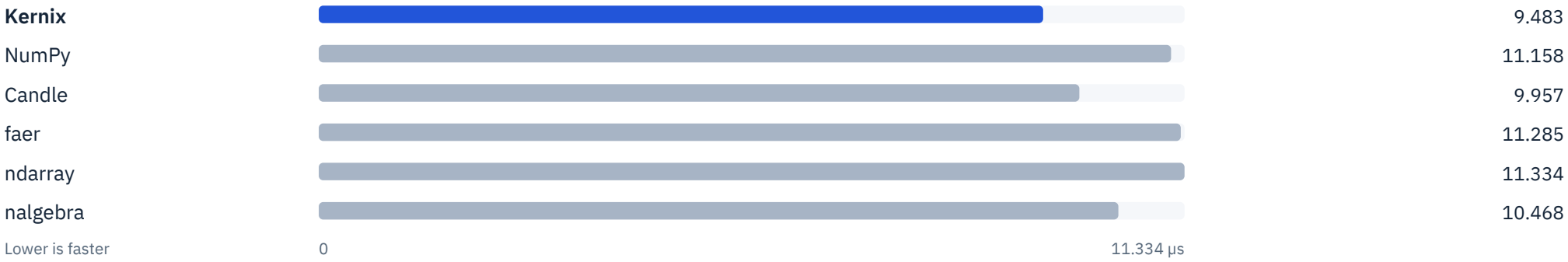
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



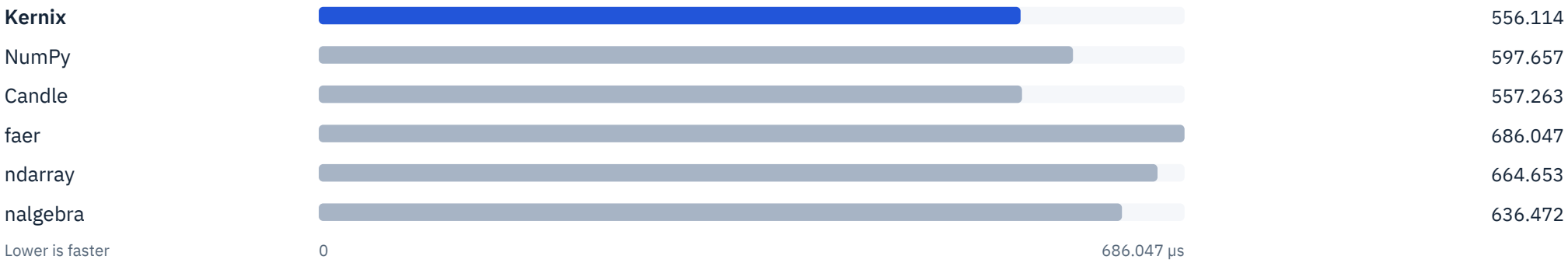
# Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



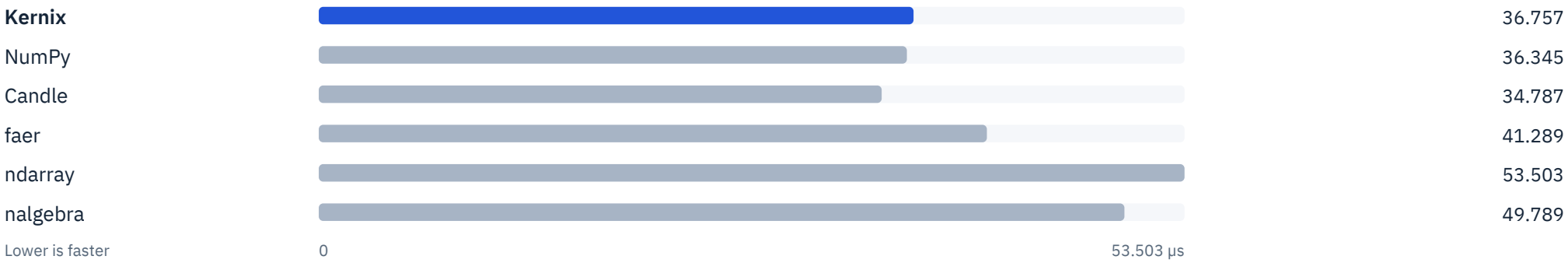
# Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Matrix multiplication

Per-call median

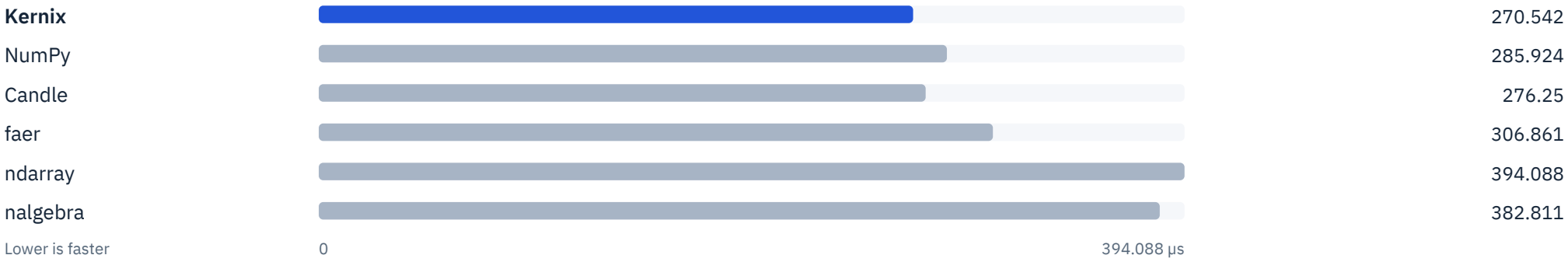
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



# Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



# Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Column-pivoted QR

Per-call median

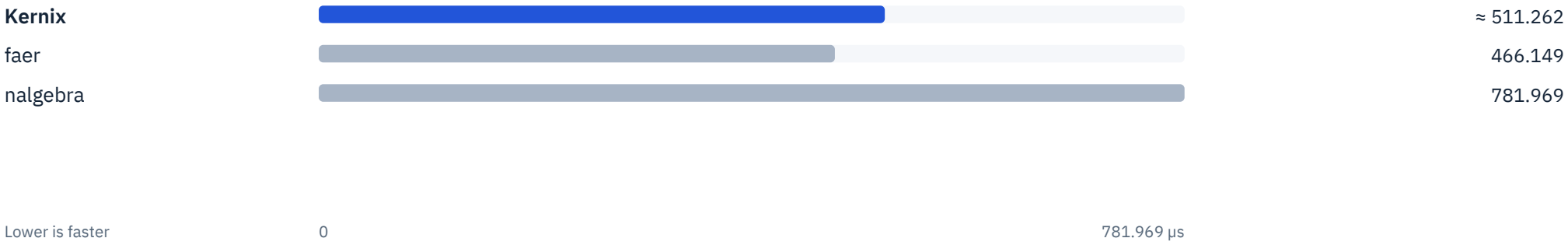
96 × 32 · f64 · 1 worker · allocated output




## Column-pivoted QR

Per-call median

256 × 64 · f64 · 1 worker · allocated output



# Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Minimum-norm least squares

Per-call median

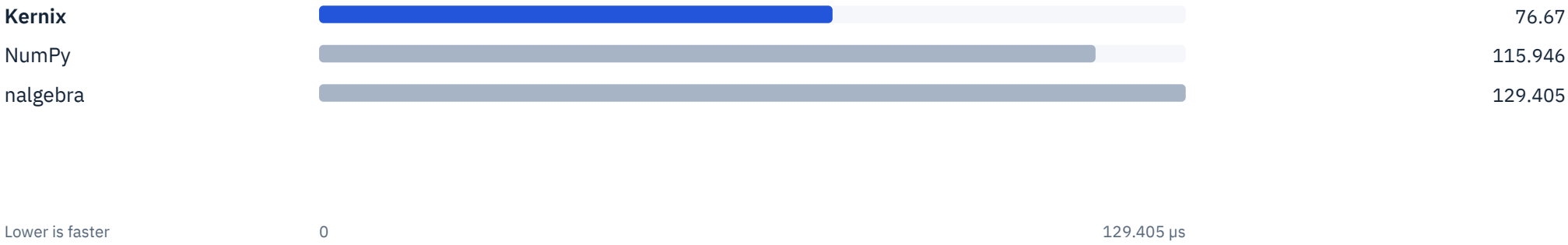
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ




## Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

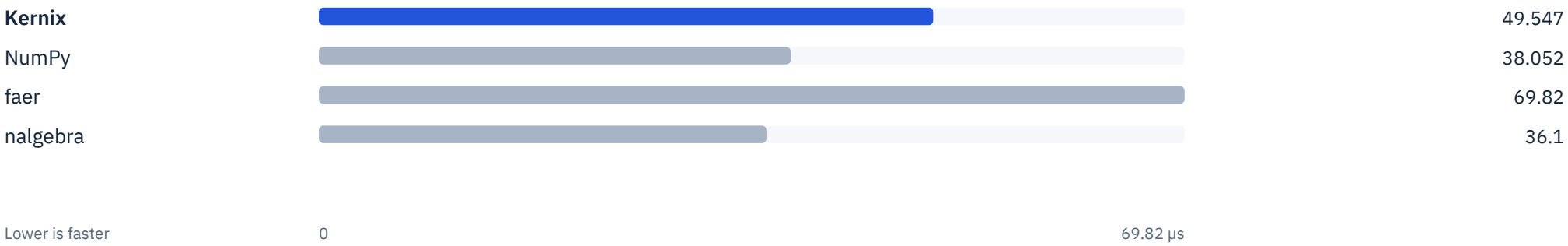


# Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

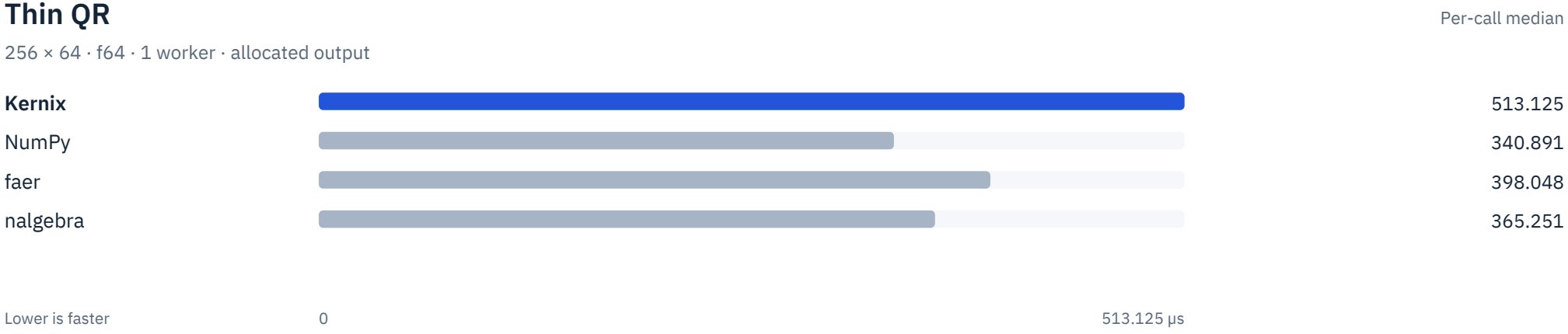
## Thin QR

96 × 32 · f64 · 1 worker · allocated output




## Thin QR

256 × 64 · f64 · 1 worker · allocated output



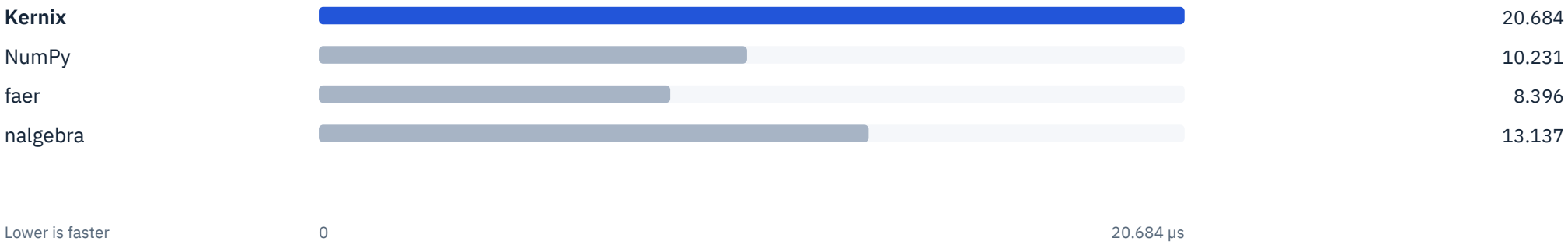
# Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky factorization

Per-call median

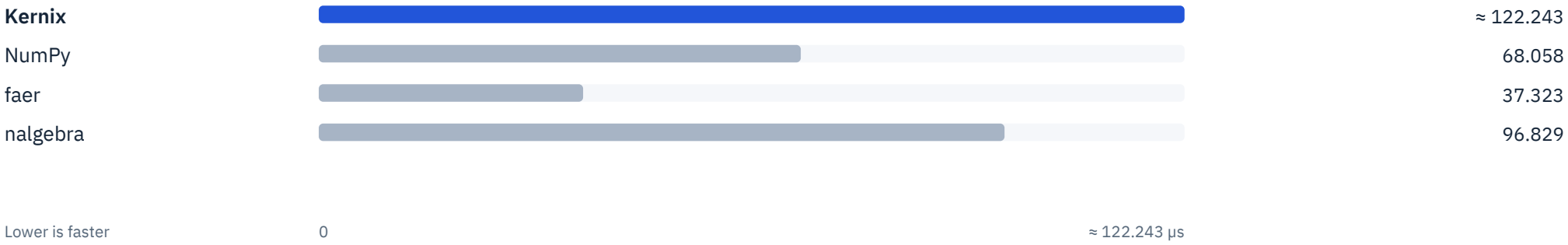
64 × 64 · f64 · 1 worker · allocated output




## Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output

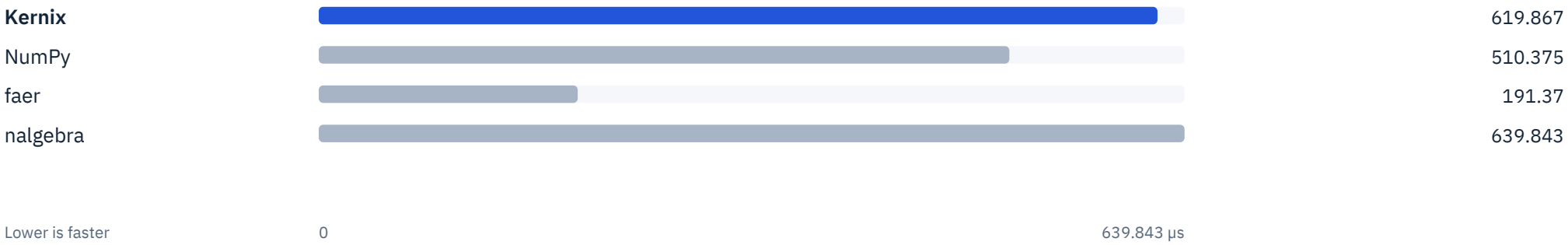


# Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



# Kernix 0.9.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Record + backward

Per-call median

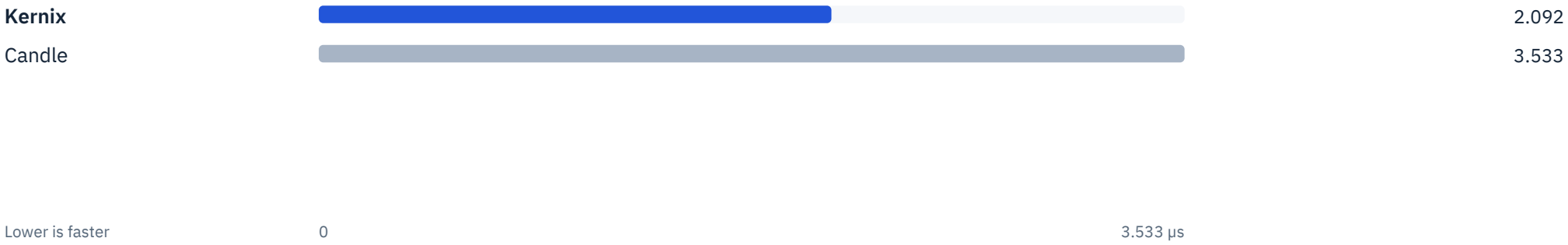
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product




## Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



# Kernix 0.9.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Model recording + forward

Per-call median

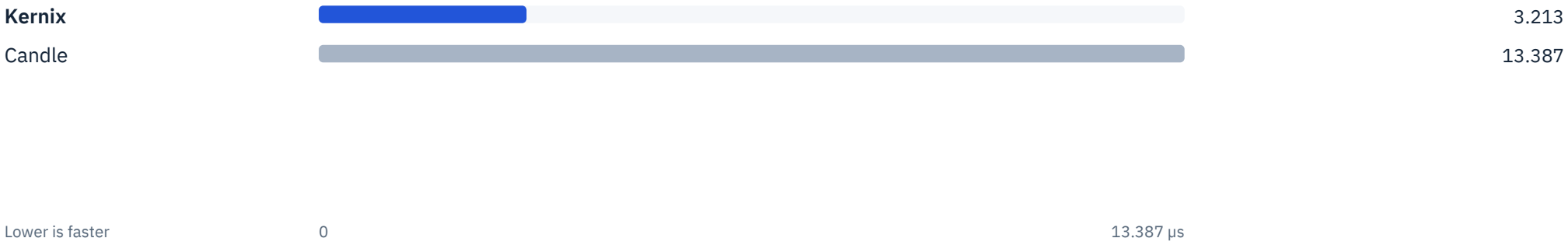
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented




## Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



# Kernix 0.9.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

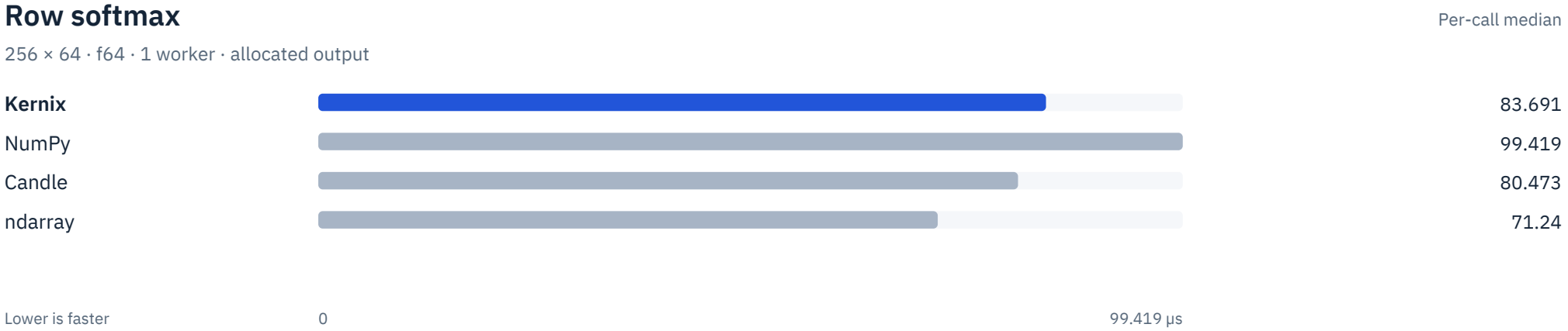


# Kernix 0.9.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Row softmax

256 × 64 · f64 · 1 worker · allocated output



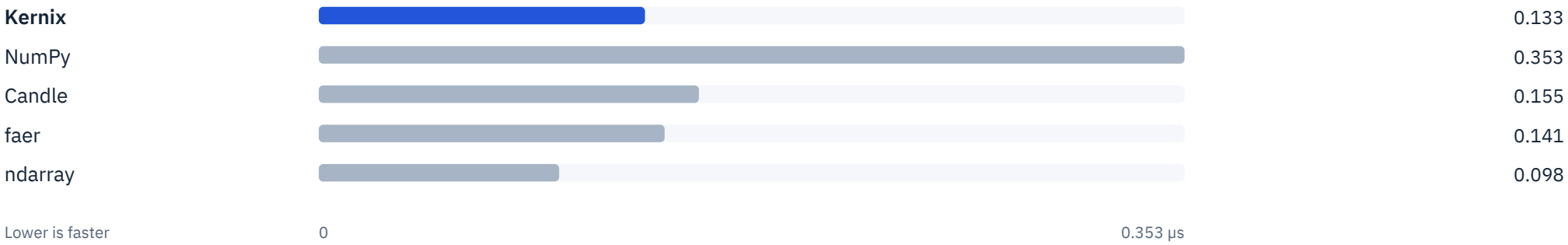
# Kernix 0.9.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

## Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output




## Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



# Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

## Cholesky factorization

64 × 64 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.9773944919635305** | 0.9235327313769752 – 1.0403771163433535 |

## Cholesky factorization

128 × 128 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.9789425417092872** | 0.9268104333538715 – 1.018511826247222 |

# Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

## Cholesky factorization

256 × 256 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0069935500152878** | 0.9932964002130146 – 1.025894234196159 |

## Thin QR

96 × 32 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0491846686820097** | 0.9656443002198263 – 1.0593508293468055 |

# Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

## Thin QR

256 × 64 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.020154379165327** | 0.9954990521073648 – 1.0457764236184073 |

## Matrix multiplication

128 × 128 · 128 × 128 · f64

**Timing direction unclear**




| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0018462734428142** | 0.9800260353103897 – 1.0123633757391148 |

# Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

## Matrix multiplication

128 × 128 · 128 × 128 · f32

**Timing direction unclear**




| | | | |
|---|---|---|---|
| CURRENT / BASELINE | **0.9867207984437114** | 95% INTERVAL | 0.9328305380091483 – 1.0138829376981746 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>32 × 32 · 32 × 32 · f64 | 1637 | 1.637 |
| **Matrix multiplication / faer**<br>32 × 32 · 32 × 32 · f64 | 1442 | 1.442 |
| **Matrix multiplication / ndarray**<br>32 × 32 · 32 × 32 · f64 | 1394 | 1.394 |
| **Matrix multiplication / Candle**<br>32 × 32 · 32 × 32 · f64 | 1420 | 1.42 |
| **Matrix multiplication / NumPy**<br>32 × 32 · 32 × 32 · f64 | 2388 | 2.388 |
| **Matrix multiplication / nalgebra**<br>32 × 32 · 32 × 32 · f64 | 1650 | 1.65 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f64 | 11158 | 11.158 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f64 | 10468 | 10.468 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f64 | 9483 | 9.483 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f64 | 11334 | 11.334 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f64 | 11285 | 11.285 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f64 | 9957 | 9.957 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f64 | 78058 | 78.058 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f64 | 92307 | 92.307 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f64 | 117273 | 117.273 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f64 | 119441 | 119.441 |
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f64 | 73171 | 73.171 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f64 | 74702 | 74.702 |
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f64 | 636472 | 636.472 |
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f64 | 597657 | 597.657 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f64 | 556114 | 556.114 |
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f64 | 557263 | 557.263 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f64 | 686047 | 686.047 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f64 | 664653 | 664.653 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f32 | 5642 | 5.642 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f32 | 6276 | 6.276 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f32 | 4885 | 4.885 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f32 | 5213 | 5.213 |
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f32 | 5708 | 5.708 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f32 | 7242 | 7.242 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f32 | 36757 | 36.757 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f32 | 36345 | 36.345 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f32 | 49789 | 49.789 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f32 | 34787 | 34.787 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f32 | 53503 | 53.503 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f32 | 41289 | 41.289 |
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f32 | 285924 | 285.924 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f32 | 306861 | 306.861 |
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f32 | 270542 | 270.542 |
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f32 | 382811 | 382.811 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f32 | 276250 | 276.25 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f32 | 394088 | 394.088 |
| **Batched multiplication / Candle**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 2103 | 2.103 |
| **Batched multiplication / NumPy**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 2865 | 2.865 |
| **Batched multiplication / Kernix**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 2160 | 2.16 |
| **Column-pivoted QR / faer**<br>96 × 32 · f64 | 71206 | 71.206 |
| **Column-pivoted QR / Kernix**<br>96 × 32 · f64 | 56637 | 56.637 |
| **Column-pivoted QR / nalgebra**<br>96 × 32 · f64 | 78306 | 78.306 |
| **Column-pivoted QR / nalgebra**<br>256 × 64 · f64 | 781969 | 781.969 |
| **Column-pivoted QR / faer**<br>256 × 64 · f64 | 466149 | 466.149 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Column-pivoted QR / Kernix**<br>256 × 64 · f64 | 511261.99999999994 | 511.26199999999994 |
| **Minimum-norm least squares / Kernix**<br>64 × 32, rank 16, 4 RHS · f64 | 49903 | 49.903 |
| **Minimum-norm least squares / NumPy**<br>64 × 32, rank 16, 4 RHS · f64 | 63224 | 63.224 |
| **Minimum-norm least squares / nalgebra**<br>64 × 32, rank 16, 4 RHS · f64 | 71614 | 71.614 |
| **Minimum-norm least squares / Kernix**<br>32 × 64, rank 32, 4 RHS · f64 | 76670 | 76.67 |
| **Minimum-norm least squares / NumPy**<br>32 × 64, rank 32, 4 RHS · f64 | 115946 | 115.946 |
| **Minimum-norm least squares / nalgebra**<br>32 × 64, rank 32, 4 RHS · f64 | 129405 | 129.405 |
| **Thin QR / faer**<br>96 × 32 · f64 | 69820 | 69.82 |
| **Thin QR / NumPy**<br>96 × 32 · f64 | 38052 | 38.052 |
| **Thin QR / nalgebra**<br>96 × 32 · f64 | 36100 | 36.1 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Thin QR / Kernix**<br>96 × 32 · f64 | 49547 | 49.547 |
| **Thin QR / faer**<br>256 × 64 · f64 | 398048 | 398.048 |
| **Thin QR / Kernix**<br>256 × 64 · f64 | 513125 | 513.125 |
| **Thin QR / nalgebra**<br>256 × 64 · f64 | 365251 | 365.251 |
| **Thin QR / NumPy**<br>256 × 64 · f64 | 340891 | 340.891 |
| **Cholesky factorization / Kernix**<br>64 × 64 · f64 | 20684 | 20.684 |
| **Cholesky factorization / nalgebra**<br>64 × 64 · f64 | 13137 | 13.137 |
| **Cholesky factorization / faer**<br>64 × 64 · f64 | 8396 | 8.396 |
| **Cholesky factorization / NumPy**<br>64 × 64 · f64 | 10231 | 10.231 |
| **Cholesky factorization / faer**<br>128 × 128 · f64 | 37323 | 37.323 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Cholesky factorization / Kernix**<br>128 × 128 · f64 | 122242.99999999999 | 122.24299999999999 |
| **Cholesky factorization / NumPy**<br>128 × 128 · f64 | 68058 | 68.058 |
| **Cholesky factorization / nalgebra**<br>128 × 128 · f64 | 96829 | 96.829 |
| **Cholesky factorization / nalgebra**<br>256 × 256 · f64 | 639843 | 639.843 |
| **Cholesky factorization / faer**<br>256 × 256 · f64 | 191370 | 191.37 |
| **Cholesky factorization / Kernix**<br>256 × 256 · f64 | 619867 | 619.867 |
| **Cholesky factorization / NumPy**<br>256 × 256 · f64 | 510375 | 510.375 |
| **Record + backward / Kernix**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 2366 | 2.366 |
| **Record + backward / Candle**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 4268 | 4.268 |
| **Model inference / Kernix**<br>batched normalized/sliced MLP · f64 | 2092 | 2.092 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Model inference / Candle**<br>batched normalized/sliced MLP · f64 | 3533 | 3.533 |
| **Model recording + forward / Kernix**<br>batched normalized/sliced MLP · f64 | 2656 | 2.656 |
| **Model recording + forward / Candle**<br>batched normalized/sliced MLP · f64 | 5141 | 5.141 |
| **Model backward / Candle**<br>prepared graph → fresh gradients · f64 | 13387 | 13.387 |
| **Model backward / Kernix**<br>prepared graph → fresh gradients · f64 | 3213 | 3.213 |
| **Parameter update / Kernix**<br>transactional SGD, fixed gradients · f64 | 552 | 0.552 |
| **Row softmax / ndarray**<br>256 × 64 · f64 | 71240 | 71.24 |
| **Row softmax / NumPy**<br>256 × 64 · f64 | 99419 | 99.419 |
| **Row softmax / Candle**<br>256 × 64 · f64 | 80473 | 80.473 |
| **Row softmax / Kernix**<br>256 × 64 · f64 | 83691 | 83.691 |

# Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Create zeros / Kernix**<br>64 × 64 · f32 | 133 | 0.133 |
| **Create zeros / faer**<br>64 × 64 · f32 | 141 | 0.141 |
| **Create zeros / ndarray**<br>64 × 64 · f32 | 98 | 0.098 |
| **Create zeros / NumPy**<br>64 × 64 · f32 | 353 | 0.353 |
| **Create zeros / Candle**<br>64 × 64 · f32 | 155 | 0.155 |
| **Create ones / faer**<br>64 × 64 · f32 | 184 | 0.184 |
| **Create ones / Kernix**<br>64 × 64 · f32 | 203 | 0.203 |
| **Create ones / Candle**<br>64 × 64 · f32 | 237 | 0.237 |
| **Create ones / NumPy**<br>64 × 64 · f32 | 1348 | 1.348 |
| **Create ones / ndarray**<br>64 × 64 · f32 | 154 | 0.154 |